

Research Results: Emoji and Political Affiliation in GitHub Repositories

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Table of Contents

- [Executive Summary](#executive-summary)
- [Methodology Overview](#methodology-overview)
- [RQ1: Repository Types and Affiliations](#rq1-what-kinds-of-github-repositories-use-politically-meaningful-emojis)
- [RQ2: Expression Methods and False Positives](#rq2-how-do-these-repositories-express-political-or-activist-support-and-how-many-cases-are-false-positives)
- [RQ3: Repository Characteristics](#rq3-what-are-the-characteristics-of-these-repositories-and-their-contributors)
- [Discussion](#discussion)
- [Summary of Key Findings](#summary-of-key-findings)
- [List of Figures](#list-of-figures)

Executive Summary

This study analyzed 4,696 GitHub repositories containing political or activist emojis to understand how developers use emoji symbols to signal political and social affiliations in their projects. Through a rigorous multi-stage validation process combining AI-based classification and manual verification, we identified 59 repositories (1.26%) with genuine political or activist affiliations.

Key Findings at a Glance

Metric	Value
Total repositories analyzed	4,696
Validated activist repositories	59 (1.26%)
Most common affiliation	UKRAINE (72.9%)
Most common expression type	Text Statement (66.1%)
False positive rate (AI classifier)	59.3%
Total stars across activist repos	556,305
Contributors statistical significance	$p = 0.0015$

Methodology Overview

Data Collection

We collected 4,696 GitHub repositories that met the following criteria:

1. Minimum popularity threshold: At least 1,000 stars
2. Emoji presence: README contains at least one political or activist-related emoji
3. Political emoji categories: Ukraine flag 🇺🇦, Palestine flag 🇵🇸, rainbow 🌈, Pride flags 🏳️‍🌈, raised fists ✊️, and other activist symbols

Validation Process

1. AI Classification (DeepSeek): Initial filtering using large language model to identify potential activist repositories
2. Manual Validation: Human review of 145 AI-flagged repositories
3. Expression Categorization: Classification of how each repository expresses political support

Expression Categories Defined

Category	Definition	Example
Text Statement	Explicit written declaration of support	"We stand with Ukraine"
Link Support	URLs to donation pages or information	"[Donate to Ukraine](...)"
Hashtag/Slogan	Use of movement hashtags	#StandWithUkraine, #BLM
Visual/Badge	Images, badges, or banners	Support badges in README
Emoji Only	Only emoji without context	Just 🇺🇦 in description

Analysis Framework

- * RQ1: Repository types and affiliation distribution
 - * RQ2: Expression methods and false positive analysis
 - * RQ3: Repository characteristics comparison (stars, contributors, ownership)
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RQ1: What kinds of GitHub repositories use politically meaningful emojis?

Overview

Among the 4,696 popular repositories that contain at least one political or activist emoji, we identified 59 repositories with a clearly validated activist affiliation (approximately 1.26% of all emoji-bearing repositories). This finding indicates that political emojis in popular GitHub projects are predominantly decorative rather than intentional political signals.

Political Affiliation Distribution

The most common affiliations were UKRAINE (n=43, 72.9%), followed by PALESTINE (n=6, 10.2%) and LGBTQ (n=5, 8.5%). The dominance of Ukraine-related repositories (72.9%) reflects the global developer community's response to the 2022 Russian invasion, demonstrating how geopolitical events influence open-source project documentation.

Affiliation	Count	Percentage	Primary Emojis
UKRAINE	43	72.9%	?? ? ? ?
PALESTINE	6	10.2%	?? ?
LGBTQ	5	8.5%	???? ???? ?
REPUBLICAN	2	3.4%	? ?
BLM	1	1.7%	?? ? ? ?
ISRAEL	1	1.7%	??
CLIMATE	1	1.7%	? ?

Technology Stack Analysis

In terms of programming languages, activist-affiliated repositories span diverse technology ecosystems:

Language	Count	Percentage
JavaScript	10	16.9%
TypeScript	8	13.6%
C#	7	11.9%
PHP	5	8.5%
Rust	4	6.8%
Python	3	5.1%
HCL	3	5.1%
Jupyter Notebook	3	5.1%
Vue	2	3.4%
Shell	2	3.4%

The prevalence of JavaScript (16.9%) and TypeScript (13.6%) suggests that web development and frontend technologies are particularly associated with activist signaling, possibly due to the documentation-heavy nature of these projects and their visibility to end users.

Repository Ownership Patterns

- * Individual Users: 38 repositories (64.4%)
- * Organizations: 21 repositories (35.6%)

Individual developers are more likely to express political stances (64.4% vs 35.6%), suggesting that personal projects provide more freedom for political expression compared to organizational repositories.

Top 10 Activist Repositories by Stars

Rank	Repository	Stars	Affiliation	Language
1	[trekhleb/javascript-algorithms](https://github.com/trekhleb/javascript-algorithms)	193,622	UKRAINE	JavaScript
2	[starship/starship](https://github.com/starship/starship)	51,811	UKRAINE	Rust
3	[Leaflet/Leaflet](https://github.com/Leaflet/Leaflet)	43,814	UKRAINE	JavaScript
4	[trekhleb/homemade-machine-learning](https://github.com/trekhleb/homemade-machine-learning)	23,746	UKRAINE	Jupyter Notebook
5	[sweetalert2/sweetalert2](https://github.com/sweetalert2/sweetalert2)	17,951	UKRAINE	JavaScript
6	[trekhleb/learn-python](https://github.com/trekhleb/learn-python)	17,400	UKRAINE	Python
7	[verdaccio/verdaccio](https://github.com/verdaccio/verdaccio)	17,236	UKRAINE	TypeScript
8	[Tyrrrz/YoutubeDownloader](https://github.com/Tyrrrz/YoutubeDownloader)	13,135	UKRAINE	C#
9	[LGUG2Z/komorebi](https://github.com/LGUG2Z/komorebi)	13,058	PALESTINE	Rust
10	[retejs/rete](https://github.com/retejs/rete)	10,887	UKRAINE	TypeScript

The top activist repository has 193,622 stars, demonstrating that political signaling occurs even in the most popular open-source projects.

Most Commonly Used Political Emojis

Emoji	Count	Primary Association
-------	-------	---------------------

?? | 31 | Ukraine
?? | 14 | Support
? | 5 | LGBTQ+
???? | 5 | Pride
?? | 5 | Russia/Anti-war
?? | 4 | Palestine
?? | 4 | Various
? | 2 | Ukraine
? | 2 | Ukraine
? | 2 | Various
???? | 2 | Trans Pride
? | 2 | Various
? | 1 | Various
? | 1 | Palestine
?? | 1 | BLM

Repository Creation Timeline

Year | Repositories Created

2010 | 1
2011 | 1
2014 | 2
2015 | 1
2016 | 6
2017 | 9
2018 | 8
2019 | 10
2020 | 9
2021 | 6
2022 | 3
2023 | 1
2025 | 2

Associated Figures

Figure 1: Distribution of Political/Activist Affiliations

[See figures in FINAL/figures/ folder]

**Figure 1. Distribution of activist affiliations among the 59 validated repositories.*

The plot shows the proportion of repositories supporting each movement, with Ukraine-related activism dominating the distribution.*

Figure 8: Repository Creation Timeline

[See figures in FINAL/figures/ folder]

**Figure 8. Timeline showing when activist-affiliated repositories were created,*

revealing temporal patterns in political signaling adoption.*

RQ2: How do these repositories express political or activist support, and how many cases are false positives?

Overview

We classified the way each repository expresses support into five categories: text statements, link-based support, slogans or hashtags, visual banners/badges, and emoji-only. This analysis reveals that most activist repositories use multiple channels to express support, with only a minority relying solely on emoji symbols.

Expression Type Distribution

Among the 59 validated activist repositories:

Expression Type	Count	Percentage	Description
Text Statement	39	66.1%	Explicit written statement of support in README
Link Support	35	59.3%	Links to donation pages, petitions, or information resources
Hashtag/Slogan	18	30.5%	Use of hashtags like #StandWithUkraine or slogans
Visual/Badge	18	30.5%	Badges, banners, or images showing support
Emoji Only	8	13.6%	Only emoji symbols without additional context

Key Insight: Text Statement is the most common support type (66.1%), indicating that developers prefer explicit, contextual methods over symbolic signaling alone.

Emoji-Only Signaling Analysis

Emoji-only signals were relatively rare, with only 8 repositories (13.6%) relying solely on activist emojis without any supporting text, link, or image. This finding is significant because it suggests that:

1. Emojis alone are insufficient for meaningful political communication
2. Context matters - developers recognize the need for explicit messaging
3. Detection systems cannot rely solely on emoji presence for classification

Multi-Expression Patterns

- * Repositories using multiple expression types: 38 (64.4%)
- * Repositories using single expression type: 21 (35.6%)

Most Common Expression Combinations:

Combination	Count
Link Support + Text Statement	12
Link Support + Text Statement + Visual/Badge	12
Hashtag/Slogan + Link Support	4
Hashtag/Slogan + Link Support + Text Statement	3
Hashtag/Slogan + Text Statement + Visual/Badge	2
Hashtag/Slogan + Link Support + Text Statement + Visual/Badge	2
Link Support + Visual/Badge	1
Text Statement + Visual/Badge	1

Validity Classification: Explicit vs Implicit Signaling

Validity Type	Count	Percentage	Description
Explicit	52	88.1%	Clear, unambiguous statement of support
Implicit	7	11.9%	Contextual inference required

The high proportion of explicit signals (88.1%) demonstrates that most developers who choose to express political support do so clearly and intentionally.

False Positive Analysis

When considering all 4,696 repositories that contain political emojis, only 59 (approximately 1.26%) were confirmed as truly activist.

AI Classifier Performance

Metric | Value
Repositories flagged by AI | 145
True positives (validated) | 59
False positives | 86
Precision | 40.7%
False positive rate | 59.3%

Common False Positive Patterns

Analysis of false positives revealed several common patterns:

- 1. Decorative emoji use: Rainbow 🌈 used for color themes, not LGBTQ+ support
- 2. Technical contexts: Emojis in commit messages or changelogs
- 3. Cultural references: Watermelon 🍉 in food/agriculture projects
- 4. Design elements: Flag emojis for internationalization features
- 5. Coincidental presence: Activist emojis appearing in unrelated contexts

Signal Strength Assessment

Political emojis in popular GitHub repositories are a weak but non-random signal of activist content. With only 1.26% of emoji-bearing repositories being truly activist, emoji detection alone is insufficient for reliable classification. Contextual analysis is essential.

Expression Patterns by Affiliation

Affiliation | Text | Link | Hashtag | Badge | Emoji Only
UKRAINE | 30 | 29 | 15 | 15 | 3
PALESTINE | 4 | 3 | 1 | 2 | 2
LGBTQ | 1 | 0 | 2 | 0 | 2
REPUBLICAN | 2 | 1 | 0 | 1 | 0
BLM | 1 | 1 | 0 | 0 | 0
CLIMATE | 1 | 1 | 0 | 0 | 0
ISRAEL | 0 | 0 | 0 | 0 | 1

Associated Figures

Figure 2: Support Type Distribution

[See figures in FINAL/figures/ folder]

*Figure 2. Support types used by activist-affiliated repositories.

Text statements and link support are the dominant methods, while emoji-only signaling is rare.*

Figure 7: Affiliation by Expression Type

[See figures in FINAL/figures/ folder]

*Figure 7. Distribution of expression types across different political affiliations,

showing how different movements prefer different communication strategies.*

RQ3: What are the characteristics of these repositories and their contributors?

Overview

We compared the 59 activist-affiliated repositories with two baselines: repositories with political emojis but no validated affiliation (4,637 repos) and repositories flagged as false positives (86 repos). This comparison reveals important differences in project characteristics.

Statistical Testing Summary

Variable	Test	Statistic	p-value	Significant (?=0.05)	Effect Size
Stars	Mann-Whitney U	146,652	0.3407	No ?	N/A
Contributors	Mann-Whitney U	169,644	0.0015	Yes ?	N/A

Repository Popularity (Stars)

Activist repositories show higher star counts compared to baseline repositories.

Group	N	Median	Mean	Std Dev	Min	Max	Total
Affiliated	59	2,864	9,429	26,116	1,012	193,622	556305
Baseline	4,637	2,450	6,018	11,501	1,000	182,605	-
False Positive	86	2,438	4,771	5,955	1,009	38,620	-

Statistical Result: The difference in star counts is not statistically significant (U = 146,652, p = 0.3407).

This suggests that political signaling is not associated with repository popularity.

Community Engagement (Contributors)

In terms of contributors, activist repositories show higher contributor counts.

Group	N	Median	Mean	Std Dev
Affiliated	59	30	113	194
Baseline	4,637	19	74	242
False Positive	86	22	48	82

Statistical Result: The difference in contributor counts is statistically significant (U = 169,644, p = 0.0015).

This suggests that activist signaling is associated with more collaborative, community-driven projects.

Owner Type Comparison

Group	Organization %	Individual User %
Affiliated	35.6%	64.4%
Baseline	46.9%	53.1%
False Positive	37.2%	62.8%

Individual developers are more likely to express political stances (64.4% vs 53.1%), suggesting personal projects provide more freedom for political expression.

Programming Language Comparison

Rank	Affiliated Repos	Baseline Repos
1	JavaScript (10)	Python (976)

2 | TypeScript (8) | TypeScript (788)
3 | C# (7) | JavaScript (527)
4 | PHP (5) | Go (238)
5 | Rust (4) | Java (206)

Associated Figures

Figure 3: Stars Comparison Across Groups

[See figures in FINAL/figures/ folder]

Figure 3. Distribution of star counts for activist-affiliated repositories compared to baseline groups.

Figure 4: Contributors Comparison Across Groups

[See figures in FINAL/figures/ folder]

Figure 4. Distribution of contributor counts, showing higher engagement in activist-affiliated repositories.

Figure 5: Language Distribution

[See figures in FINAL/figures/ folder]

Figure 5. Primary programming languages comparison between affiliated and baseline repositories.

Figure 6: Owner Type Distribution

[See figures in FINAL/figures/ folder]

Figure 6. Owner types showing the proportion of individual vs organizational ownership.

Figure 9: Stars vs Contributors Relationship

[See figures in FINAL/figures/ folder]

Figure 9. Scatter plot showing the relationship between repository popularity and community size.

Discussion

Key Findings

1. Political Emoji Usage is Rare but Meaningful

Only 1.26% of repositories containing political emojis actually have genuine activist affiliations. This indicates that political emojis are predominantly used for decorative purposes in the developer community. However, when political signaling does occur, it is typically explicit and intentional (88.1% explicit vs 11.9% implicit).

2. Ukraine Dominates Activist Signaling

With 72.9% of activist repositories supporting Ukraine,

the 2022 Russian invasion clearly had a significant impact on the open-source community. This reflects broader global solidarity movements and the developer community's response to geopolitical crises.

3. Multi-Modal Expression is Preferred

Developers rarely rely on emoji-only signaling (13.6%). Instead, 64.4% use multiple expression methods, combining text statements, links, badges, and hashtags for comprehensive messaging.

4. Community Engagement Matters

The statistically significant difference in contributor counts ($p = 0.0015$) suggests that activist signaling is associated with more collaborative projects. This may indicate that community-oriented projects are more likely to take public stances on social issues.

Limitations

1. Sample bias: Only repositories with 1,000+ stars were analyzed
2. Temporal scope: Data represents a snapshot; political movements evolve
3. Classification accuracy: AI classifier has 59.3% false positive rate
4. Cultural context: Emoji interpretations vary across cultures
5. Language limitations: Analysis focused on English README content

Implications for Research and Practice

1. For researchers: Emoji detection alone is insufficient for political content classification
 2. For platforms: Political signaling in open source deserves consideration in moderation policies
 3. For developers: README badges provide effective channels for social advocacy
 4. For organizations: Political positioning in open source is increasingly visible
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Summary of Key Findings

RQ1: Repository Types and Affiliations

- * 59 repositories (1.26%) contain genuine activist affiliations
- * UKRAINE dominates with 72.9% of affiliated repositories
- * Primary languages: JavaScript, TypeScript, C#
- * Individual users own 64.4% of activist repositories
- * Total stars across activist repos: 556,305

RQ2: Expression Methods and False Positives

- * Text Statement is the most common expression type (66.1%)
- * Only 13.6% rely on emoji-only signals
- * 64.4% use multiple expression methods
- * 88.1% explicit vs 11.9% implicit signaling

- * 59.3% false positive rate among AI predictions
- * Political emojis are a weak signal: only 1.26% of emoji repos are truly activist

RQ3: Repository Characteristics

- * Affiliated repos have higher median stars (2,864 vs 2,450)
 - * Affiliated repos have higher median contributors (30 vs 19)
 - * Stars difference is not statistically significant ($p = 0.3407$)
 - * Contributors difference is statistically significant ($p = 0.0015$)
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List of Figures

Figure	Filename	Research Question	Description
Figure 1	figure1_affiliation_distribution	RQ1	Distribution of political affiliations
Figure 2	figure2_support_type_distribution	RQ2	Expression type distribution
Figure 3	figure3_stars_comparison	RQ3	Stars comparison across groups
Figure 4	figure4_contributors_comparison	RQ3	Contributors comparison
Figure 5	figure5_language_distribution	RQ3	Programming language distribution
Figure 6	figure6_owner_type_distribution	RQ3	Owner type comparison
Figure 7	figure7_affiliation_by_expression	RQ2	Expression by affiliation
Figure 8	figure8_creation_timeline	RQ1	Repository creation timeline
Figure 9	figure9_stars_contributors_scatter	RQ3	Stars vs contributors scatter
